

NITHIN R C

Associate Consultant

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Summary

AI Solutions Expert with 3 years' experience in building production grade **autonomous agent** systems with intelligent, self-healing automation. Expert in multi-agent orchestration (**LangGraph, LangChain, Azure OpenAI**) with production **RAG** pipelines, guardrail frameworks, and **MCP** tool-calling architectures. Delivered agentic AI solutions across **Healthcare, Finance** and **Energy** sectors achieving measurable business impact at enterprise scale.

Experience

Infosys

AI Engineer | Solutions Architect

03/2025 – Present

- **AI Integration & Team Enablement:**

Drove AI solution integration across cross-functional teams, researched, and recommended **emerging AI technologies** for project innovation, and conducted training sessions on advanced AI development methodologies to upskill team members.

- **Agentic AI Powered-Self-Healing AI Pipeline for Autonomous BI Operations:**

Lead the team in building an LLM-powered, self-healing agentic system that fully automated the end-to-end BI refresh pipeline - a process previously handled manually by a dedicated operations team involving data quality validation, report refreshes across Power BI and Tableau, failure triaging, and stakeholder notifications. The system uses **Lang Graph** and **Azure OpenAI (GPT-40)** to autonomously make decisions at each stage - resolving configuration, running Colibra DQ checks as a quality gate, triggering conditional BI refreshes, and sending intelligent notifications. Integrated RAG-based auto-remediation using Chroma DB over operational runbooks, enabling the agent to diagnose and self-heal common failures without human intervention, cutting mean-time-to-resolution by **~70% and freeing up ~15-20 hours/week** of manual effort.

- **Autonomous Multi-Agent AI System for Pharmaceutical Marketing:**

Engineered **agentic AI** architecture using the **Lang Graph** framework and **OpenAI GPT-4** to orchestrate intelligent email campaigns targeting over 12,000 doctors across more than 28 specialties. Built a full-stack platform (Fast API backend, React dashboard) with LLM-powered content generation and dual email provider orchestration achieving a 42% open rate (2x the industry average), a 15.7% CTR, and a 75% reduction in campaign setup time, demonstrating production-grade expertise in **agent-based systems**, and **LLM orchestration**.

- **AI Researcher:**

Conducted foundational research exploring applications of AI in **quantum computing** and **robotics**, evaluating emerging trends, algorithms, and integration potential.

Sonata Software

Gen AI Developer

03/2023 - 03/2025

- **Advanced Generative AI Expertise:** Designed, trained and fine-tuned state-of-the-art generative AI models (GPT, DALL-E) using LangChain, TensorFlow, PyTorch, and Hugging Face. Built end-to-end AI pipelines from data preprocessing to deployment for chatbots, content creation tools, and personalized recommendation systems.

- **Responsible AI Leadership - Harmoni.AI:** Led the Responsible AI framework integration into Sonata's Harmoni.AI platform, embedding ethical AI practices, including bias mitigation, transparency, and accountability, while enabling faster time-to-market with secure, scalable AI-driven insights for enterprise clients.

- **Azure AI Solutions Architecture:** Architected and optimized enterprise AI solutions leveraging Azure Cognitive Services (Document Intelligence, Computer Vision, Face API, LUIS, Text Analytics, Speech Services), and Azure OpenAI integration to automate and enhance client applications.

Skills

AgenticAI

Cloud Computing

Generative AI

AI/ML

Prompt Engineering

Robotics

Quantum computing

Retrieval-augmented generation

Model Context Protocol (MCPs)

Transformers

Neural networks

Computer Vision

Azure AI Foundry

Azure OpenAI

LangGraph-LangChain

Python

Snowflake.

Docker

Vector Databases

Apache Airflow

Terraform

Projects

Agentic AI Powered-Self-Healing AI Pipeline for Autonomous BI Operations:

Tech Stack: LangGraph | Python | Azure Data Factory | RAG-MCP | Power BI 08/2025

- Productivity & Business Impact:** Eliminated ~15-20 hours/week of manual operations effort by fully automating DQ validation, conditional BI refreshes, failure diagnosis, and stakeholder notifications. **RAG-powered** auto-remediation reduced mean-time-to-resolution from hours to minutes for known failure patterns. Zero-config onboarding-adding a new pipeline requires only a Snowflake database insert, not code changes. Every agent decision and tool call is audit-logged to Snowflake for enterprise compliance.
- Flow & Key Technical Details:** ADF webhook triggers the orchestrator with agent resolves config from Snowflake → runs Collibra DQ checks as a quality gate → refreshes Power BI/Tableau reports → logs execution → sends event-driven notifications on failure, queries a ChromaDB vector store over runbooks and auto-remediates (confidence ≥ 0.70) or escalates. Built on a **LangGraph** StateGraph with a **ReAct-style** agent loop (observe → decide → act), supporting both single-agent and multi-agent supervisor modes with specialist sub-agents. Includes a 5-layer guardrail chain (injection detection, hallucination validation, policy enforcement, token budget, rate limiting), semantic caching for **LLM** cost reduction, smart context-window pruning (~120K token budget), **MCP** tool server (JSON-RPC 2.0) exposing 15+ tools, and full observability via OpenTelemetry + LangSmith.

Autonomous Multi-Agent AI System for Pharmaceutical Marketing:

Tech Stack: LangGraph | FastAPI | Python 03/2025

- Productivity & Business Impact:** Engineered Agentic AI architecture using **LangGraph** framework to orchestrate intelligent email campaigns targeting 12,000+ doctors across 28+ specialties achieving 42% open rates (2x industry average), 15.7% CTR, and 75% reduction in campaign setup time through AI-driven personalization and optimization.
- Flow & Key Technical Details:** Built production-grade system with **FastAPI** backend (15+ RESTful endpoints, SQLAlchemy ORM), React analytics dashboard, **OpenAI GPT-4** integration for context-aware medical content generation, and dual email provider orchestration with NPI-based doctor intelligence and engagement prediction. Developed 3-stage drip campaigns with behavioural triggers, real-time analytics tracking 10+ metrics, HIPAA/CAN-SPAM compliant architecture with automated opt-out management—demonstrating expertise in agent- based systems, **LLM** orchestration, and healthcare domain applications.

Personal Projects & POC's

Tech Stack: AutoGen | LLMs | RAG | MCP | LangGraph | Python(pandas/numpy/matplotlib/OpenCV/TensorFlow/PyTorch) 08/2024

- AI Multi-Agent Financial Analysis System:** Multi-agent AI system leveraging **AutoGen** & Gemini LLM to automate financial research through coordinated web scraping, stock analysis and automated report generation with visualizations.
- Document Search Bot:** Designed and implemented a Generative AI-powered bot using Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs) to enable seamless document uploads and intelligent search capabilities.
- Medical Assistant:** Built a Generative AI-based bot leveraging RAG and LLMs to assist patients by providing accurate and timely responses to their medical queries.
- Geetha-GPT:** Developed a specialized bot focused on the Bhagavad Gita, utilizing Generative AI to provide insights and answers related to the sacred text.
- Eye Pupil Movement Based Pin Authentication System:** Developed a security authentication system leveraging AI

Education

Don Bosco Institute of technology

B.E. in Information science Engineering - 8.0 CGPA 08/2018 - 08/2022

Awards

& DNA Award 'RUN FOR IT' Ğ PEER 360 Award 'The Collaborator' * PEER 360 Award 'The Role Model'

Certifications

Microsoft AI 900 Infosys Certified AI Professional Q&A ltd Certification on Azure OpenAI Advance

Languages

English Native Hindi Proficient Kannada Proficient Telugu Proficient